

Science

- 1) Active transport involves all of the above.
- 2) Water which is absorbed by roots is transported to aerial parts of the plants through xylem.
- 3) During transpiration there is loss of water.
- 4) Root hairs are both a and b.
- 5) Which of the following process requires energy?
Active transport.
- 6) The wall of human heart is made of all of the above.
- 7) Which is the correct sequence of blood flow?
atrium $\rightarrow$ Ventricles $\rightarrow$ arteries $\rightarrow$ vein.
- 8) A patient with blood group O was injured in an accident and has blood loss which group of blood

should be used by doctor for transfusion?
O group.

- 9) Heart of heart is called SA node.
- 10) Which one of the following shows correct composition of blood.
Blood - plasma + RBC + WBC + Platelets.
- 11) Transpiration involves evaporative loss of water from aerial plants.
- 12) Water enters into the root hair cell through semi-permeable membrane plasma.
- 13) Part of the root that absorbs water from the soil is Root hairs.
- 14) Normal Blood pressure is $120\text{ mm}/80\text{ mmHg}$.
- 15) The normal human heart beat is about 72-75 time per minute.

iii) Match

- 1) Symplastic pathway - Plasmodesmata
 - 2) Transpiration - leaf.
 - 3) Osmosis - pressure gradient.
 - 4) Root pressure - pressure in Xylem.
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- 1) Leukemia - Blood cancer.
 - 2) Platelets - Thrombocytes.
 - 3) Monocytes - phagocytes.
 - 4) Leucopenia - Decrease in leucocytes.
 - 5) AB Blood group - Absence of antibody.
 - 6) O blood group - Absence of antigen.
 - 7) Eosinophil - Allergic conditions.
 - 8) Neutrophils - Inflammation.

iv True or false:

- 1) The phloem is responsible for the translocation of food. True.
- 2) Plants lose water by the process of transpiration. True.
- 2) The forms of sugar transported through the phloem is sucrose. True.
- 4) In apoplastic movement the water travels through the cells cell membrane and enter the cell. False.
In symplastic membrane movement the water travels through the cell membrane and enter the cell.
- 5) When guard cells lose water the stoma opens. False.
When guard cells lose water the stoma closes.
- 6) Initiation and stimulation of heart beat take place by nerves. False, Initiation and stimulation of heart beat take place by muscles fibres.
- 7) All veins carry deoxygenated blood. False, Alls veins carry deoxygenated blood except pulmonary veins.
- 8) WBC defend the body from bacterial and viral infections. True.
- 9) The closure of the mitral and tricuspid valves at the start of the ventricular systole produces the first sound "LUBB".

Answer in word or sentence:-

- 1) Name two layered protective covering of human heart

The heart is enclosed in a double walled sacs called pericardium.

- 2) What is the shape of RBC in human blood?
Biconcave and disc shaped.

- 3) Why is the colour of the blood red?

The RBCs impart red colour to the blood. Due to the presence of respiratory pigments haemoglobin.

- 4) Which kind of cells are found in the lymph?
White Blood Corpuscles.

- 5) Name the heart valve associated with the major arteries leaving the ventricles.
Semilunar valve

- 6) Mention the artery which supplies blood to the heart muscle.
Coronary artery.

Assertion and Reason:-

Assertion :- RBC plays an important role in the transport of respiratory gases.

Reason :- RBC do not have cell organelles and nucleus

If both A and R are true and R is correct explanation of A.

- 2) Assertion :- Persons with AB blood group are called an universal recipients, because they can receive blood from all groups.
Reason :- Antibodies are absent in persons with AB Blood group.

If both A and R are true and R is correct explanation of A.

English

- 1) Some crickets have four legs and some have two. Elucidate the statement from the poet point of view?

Poem :- The Ant and the cricket

Source :- Adapted from Aesop's fables

Theme :- Some crickets have four legs and some have two

This narrative poem, the poet brings out the idea of hard work and pre-planning that serves the moral to the readers. "Some crickets have four legs and some have two" is the last line of the poem that serves the moral to the reads. The poem is about a wise, hardworking and a careless lazy cricket. The young, silly cricket enjoyed singing all summer and spring not worry about the winter, It had approach an ant for food and shelter. During the winter, it suffered a lot with out food and shelter. If he had

stored some grains in the winter, he would not begged the ant in the winter. Likewise some people enjoyed the life when they are capable of working. They never plan or save anything for their future. And they are feel for it later. The poet indicated that ^{they} are the two legged cricket. He wants such people to work hard and save for their future.

Moral : Work hard today , gets benefits tomorrow

Every Creature will
Every Creature will
Every Creature will

Science

Short Answers :

1) What causes the opening and closing of guard cells of stomata during Transpiration?
* The opening and closing of the stomata is due to the change of turgidity of the guard cells

* When water enters into the guard cells they became turgid and the stoma opens.

* When the guard cells lose water, it became of flaccid and the stoma closes.

2) What is cohesion ?

The force of attractive between molecules of water is called cohesion.

3) Trace the pathway followed by water molecules from the time it enters plant root to the time it escape into the atmosphere from a leaf.

Root pressure → Cohesion → Transpiration pulls → Capillary Action → Adhesion → Transpiration.

4) What would happen to the leaves of a plant that transpires more water than its absorption in the roots?

* When the leaves of a plant transpires more water than its absorption, it resulting in excessive transpiration.

* This leads to the loss of necessary moistures from the plant body resulting in wilting or drying of a leaves.

* It may also lead to death of a plant.

5) Describe the structure of heart and working of the human heart.

structure: * The human heart is made up of Cardiac muscles and is enclosed by the pericardium.

* It is four chambered. The two upper chambers are called auricles and the two lower chambers are called ventricles.

* The tricuspid is located between the right auricles and the right ventricles.

* The bicuspid is located between the left auricles and the left ventricles.

Working:

The heart is a muscular pumping organs that pumps out blood into the blood pressure.

6) Why is the circulation in man referred to as double circulation

* In man, the blood circulates twice through the heart in one complete cycle. So it is called as a double circulation.

* In double circulation the oxygenated blood does not mix with the deoxygenated blood.

7) What are heart sounds? How are they produced?

* The rhythmic closure and opening of the valves of the heart cause heart sound.

* The first sound LUBB is of longer duration and is produced by closure of the tricuspid and the bicuspid valve after the beginning of the ventricular systole.

* The second sound DUBB is of shorter duration and is produced by closure of the semilunar valves at the end of the ventricular systole.

Corpuscles
Corpuscles
Corpuscles